COMPARISON REPORT: USB-PD MODULE

1. Executive Summary

This report evaluates three PCB fabrication houses (JLCPCB, PCBWay, NextPCB) for the USB-PD Module prototype run. The objective was to minimize lead times and costs while maintaining the highest manufacturing standards (including X-Ray inspection for BGA components).

Decision: JLCPCB was selected as the primary vendor. Although a minor strategic redesign was required to bypass a 19-day supply chain bottleneck on a specific WLCSP component (FPF1048BUCX), this pivot reduced the overall turnaround time by nearly a month and resulted in the lowest total project cost.

2. Detailed Vendor Analysis

A. JLCPCB (Selected Vendor)

To leverage JLCPCB's rapid 2-5 day turnaround, we performed a footprint redesign to replace the FPF1048BUCX load switch (which faced MOQ restrictions and delays) with a readily available Texas Instruments alternative (TPS22950CDDCR). Several passives were also swapped to match their immediate local stock.

Component Swaps / Replacements:

Designator	Original Part	Replacement Part (In-Stock)	Status
U1	FPF1048BUCX (WLCSP)	TPS22950CDDCR (SOT-23-6)	Redesigned & Replaced
J4	USB4151-GF-C	2188470001 (Molex)	Replaced
TP3-TP6	SMD_0603	RCUCTE	Replaced
R32, R33	10 Ohms	RC0402FR-0710RL	Replaced
R36	200k Ohms	RC0603JR-07200K	Replaced
R37, R39	649k Ohms	RC0603FR-07649K	Replaced

R38, R40	120k Ohms	RC0603JR-07120K	Replaced

- **Lead Time:** 2 to 5 production days.
- **Pricing (5 Units):** * PCB & Assembly: $314.53 *(Includes Epoxy-Filled Via-in-Pad upgrade)*
  - Shipping: $13.20
  - **Total Cost: $327.73**

B. PCBWay

PCBWay was capable of manufacturing the original design without footprint modifications. However, they faced critical inventory shortages for key components, requiring significant procurement delays.

Component Shortages & Sourcing Delays:

Designator	Component	Footprint	Expected Sourcing Delay
U11	MAX77958EWV+T	WLP30	**7 to 10 Workdays**
TP3-TP6	RCUCTE	SMD_0603	**7 to 10 Workdays**

- **Lead Time:** 16 to 17 production days + 7 to 10 days sourcing delay (Approx. 3-4 weeks total).
- **Pricing (5 Units):** * PCBs: $217.07
  - Assembly: $113.59
  - **Estimated Total: $330.66** *(Excludes Shipping)*

C. NextPCB

NextPCB also supported the original design but exhibited the most severe supply chain issues and the longest lead times of all evaluated vendors.

Component Shortages & Sourcing Delays:

Designator	Component	Expected Sourcing Delay
U16	FPF1048BUCX	**7 to 18 Days**
U11	MAX77958EWV+T	**4 to 7 Days**
U17	TPS61253AYFFT	**4 to 7 Days**

S4	JS202011JCQN	**4 to 7 Days**
R32, R33	10 Ohms	**7 to 18 Days**
R36, R37, R39	200k, 649k Ohms	**7 to 18 Days**
R38, R40	120k Ohms	**7 to 18 Days**
TP3-TP6	RCUCTE	**Out of Stock** (No availability)
J4	USB4151-GF-C	Replaced with 2188470001 (4-7 Days delay)

- **Lead Time:** 25 production days + 20 days procurement delay (Approx. 45 days total).
- **Pricing (5 Units):** * PCBs: $179.14
  - Procured Materials: $41.33
  - Assembly: $250.08 *(Real cost without one-time promo)*
  - **Estimated Total: $470.55** *(Excludes Shipping)*

3. Final Comparison Matrix

Vendor	Design Changes Required?	Est. Total Lead Time	Est. Total Cost (5 Units)	Verdict
JLCPCB	**Yes (Strategic)**	**2 to 5 Days**	**$327.73** *(Includes Shipping)*	**SELECTED**
PCBWay	No	23 to 27 Days	$330.66 *(Plus Shipping)*	Rejected (Too slow)
NextPCB	No	~ 45 Days	$470.55 *(Plus Shipping)*	Rejected (Too slow & expensive)